

# Backend .NET Internship

Week 4 — Day 4

## Postman Validation Testing

Create & Update Book Endpoints

Endpoint 1

**POST /api/books**

Endpoint 2

**PUT /api/books/{id}**

**7**

Create Tests

**7**

Update Tests

**14**

Total Tests

# Section 1

## Create Book Endpoint Validation

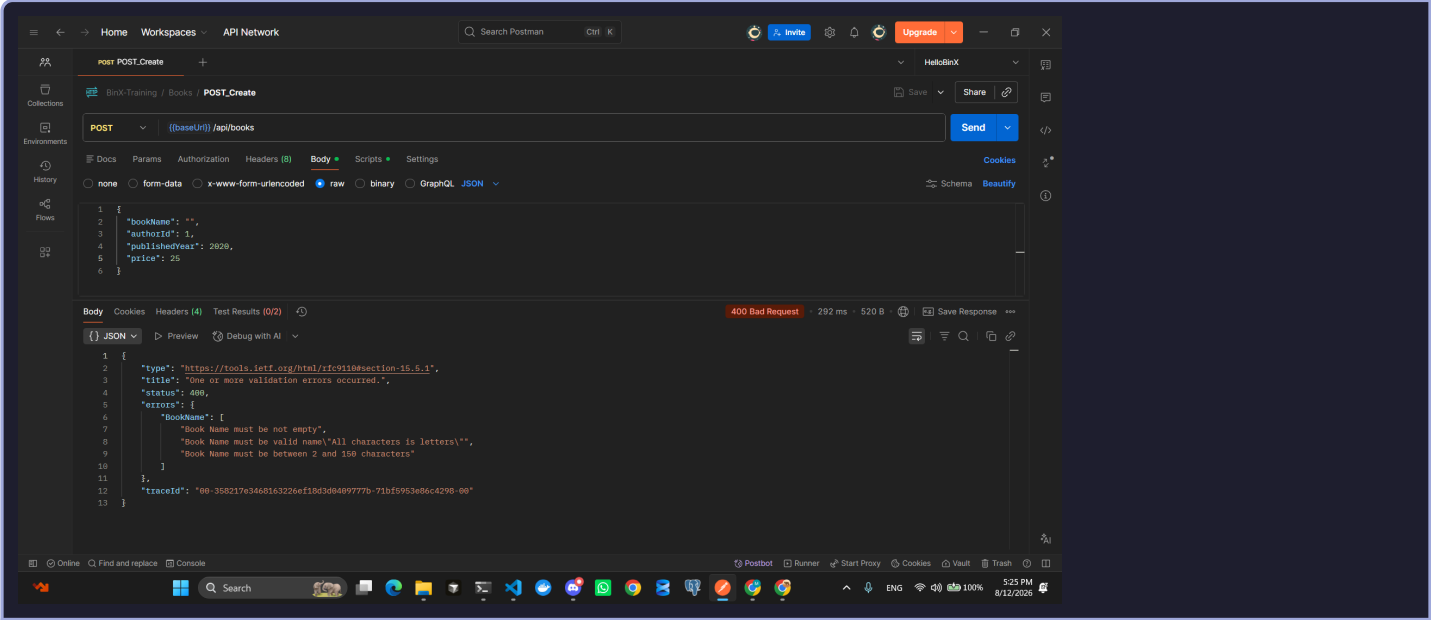
POST

/api/books

The following test cases validate the request body constraints enforced by the Create Book endpoint. Each case sends an intentionally invalid (or valid) payload and captures the server's response.

| Test Case 1       | Empty Book Name                                                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description       | Sending an empty string as the book name to verify the server rejects blank input.                                                                                                       |
| Field Tested      | bookName                                                                                                                                                                                 |
| Input Value       | "" (empty string)                                                                                                                                                                        |
| HTTP Status       | 400 Bad Request                                                                                                                                                                          |
| Validation Result | Server returned three validation errors: "Book Name must be not empty", "Book Name must be valid name!All characters is letters!", and "Book Name must be between 2 and 150 characters". |

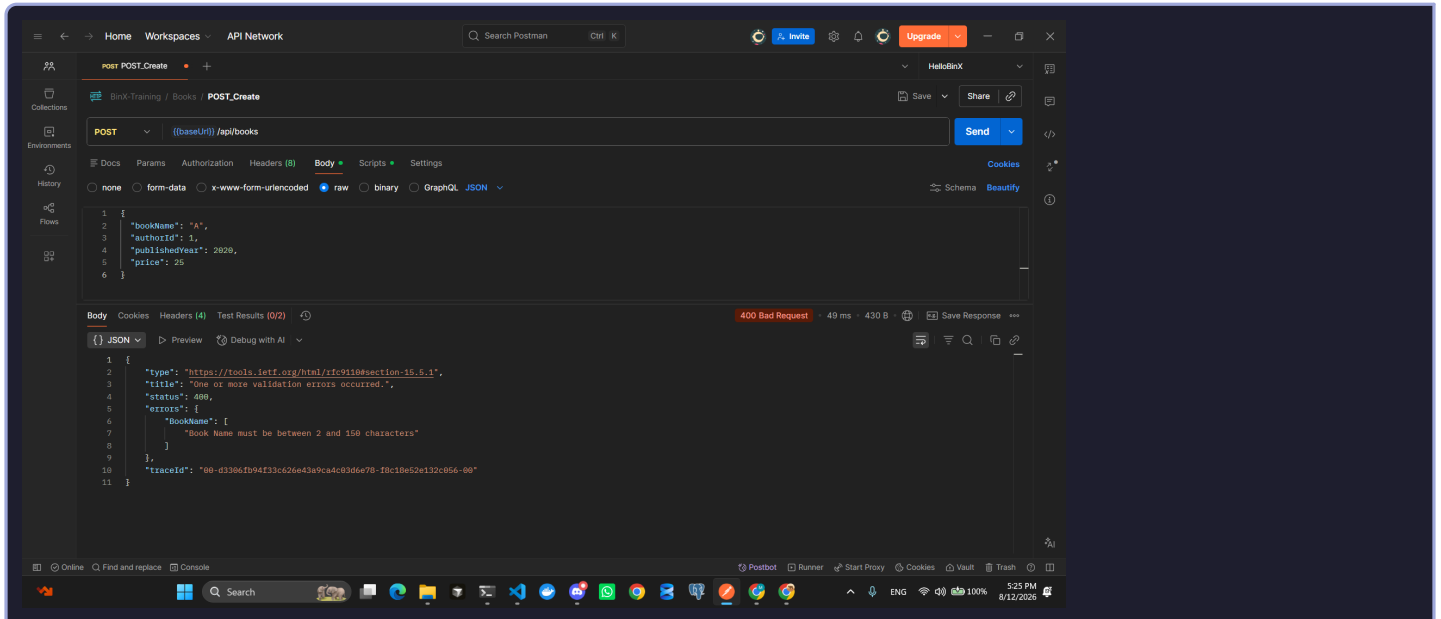
### ■ Postman Screenshot



## Test Case 2 Book Name Too Short (1 character)

|                   |                                                                               |
|-------------------|-------------------------------------------------------------------------------|
| Description       | Sending a single-character book name to verify the minimum-length constraint. |
| Field Tested      | bookName                                                                      |
| Input Value       | "A" (1 character)                                                             |
| HTTP Status       | <b>400 Bad Request</b>                                                        |
| Validation Result | Server returned: <b>"Book Name must be between 2 and 150 characters"</b> .    |

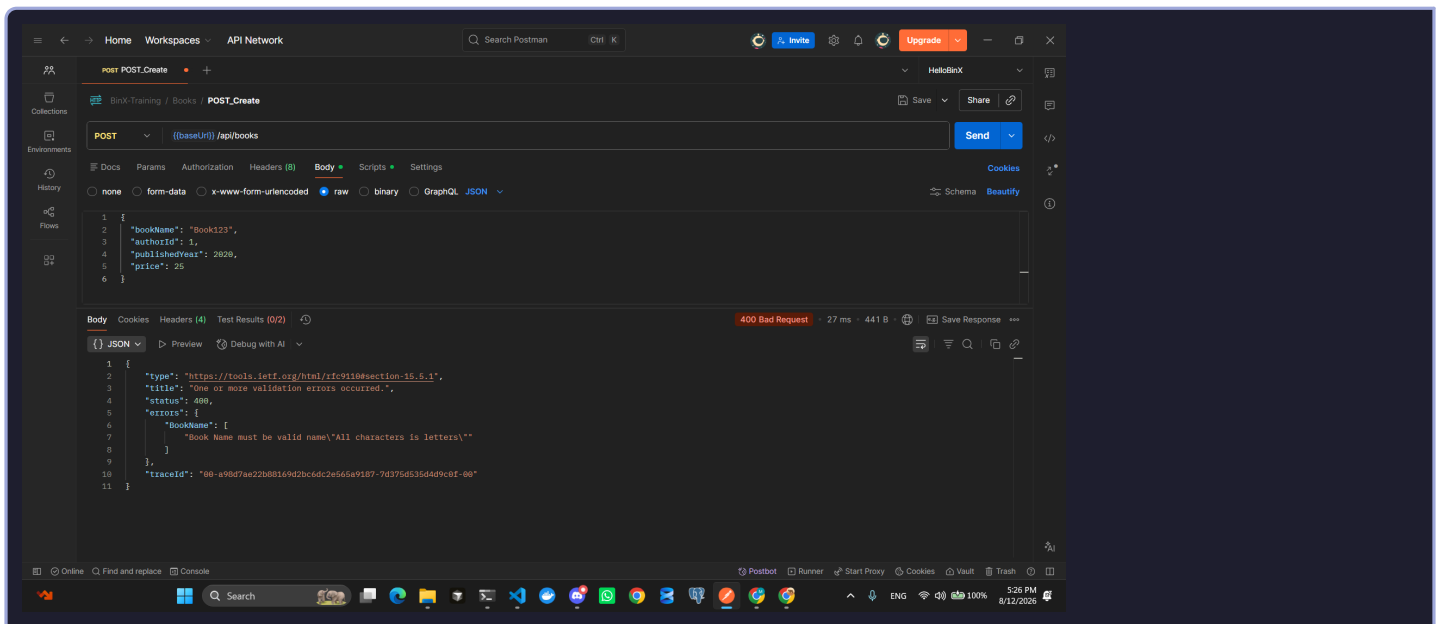
### ■ Postman Screenshot



### Test Case 3 Book Name Contains Invalid Characters

|                   |                                                                                                        |
|-------------------|--------------------------------------------------------------------------------------------------------|
| Description       | Sending a book name with alphanumeric characters (e.g., digits) to verify the letters-only constraint. |
| Field Tested      | bookName                                                                                               |
| Input Value       | "Book123" (contains digits)                                                                            |
| HTTP Status       | <b>400 Bad Request</b>                                                                                 |
| Validation Result | Server returned: "Book Name must be valid name!All characters is letters!"".                           |

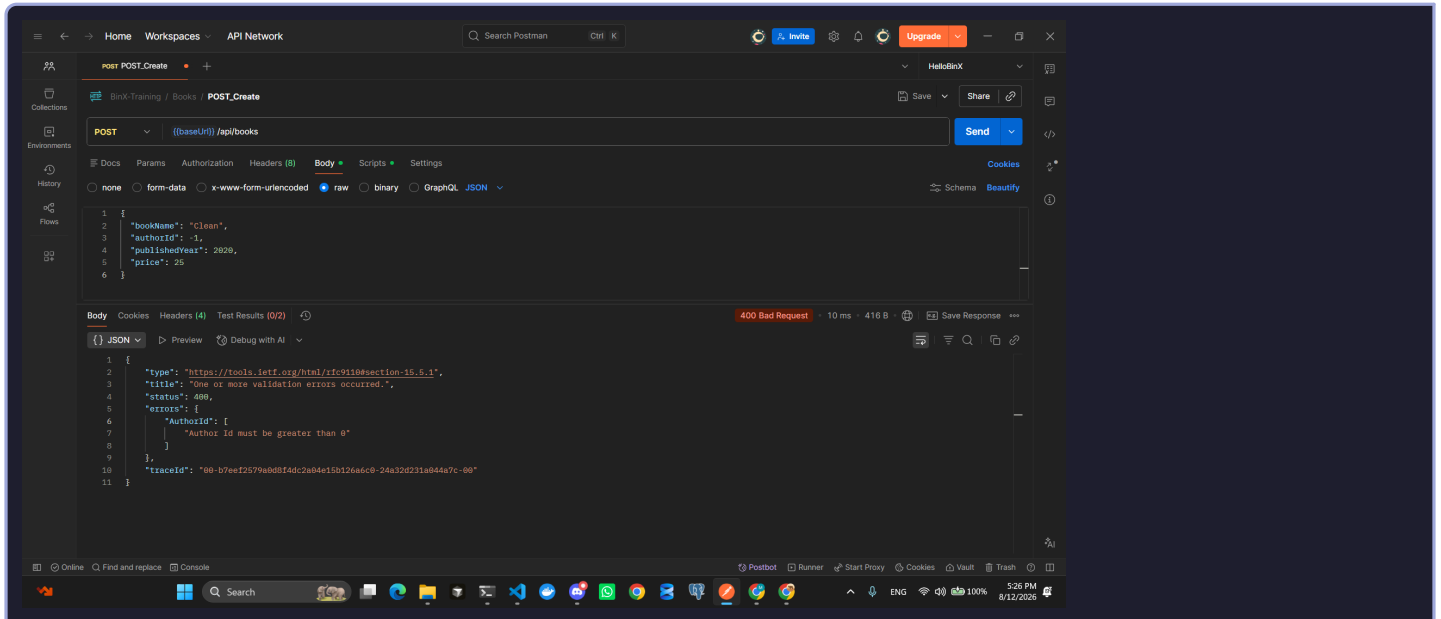
#### ■ Postman Screenshot



#### Test Case 4 Invalid Author ID (negative value)

|                   |                                                                               |
|-------------------|-------------------------------------------------------------------------------|
| Description       | Sending a negative authorId to verify that the ID must be a positive integer. |
| Field Tested      | authorId                                                                      |
| Input Value       | -1                                                                            |
| HTTP Status       | <b>400 Bad Request</b>                                                        |
| Validation Result | Server returned: "Author Id must be greater than 0".                          |

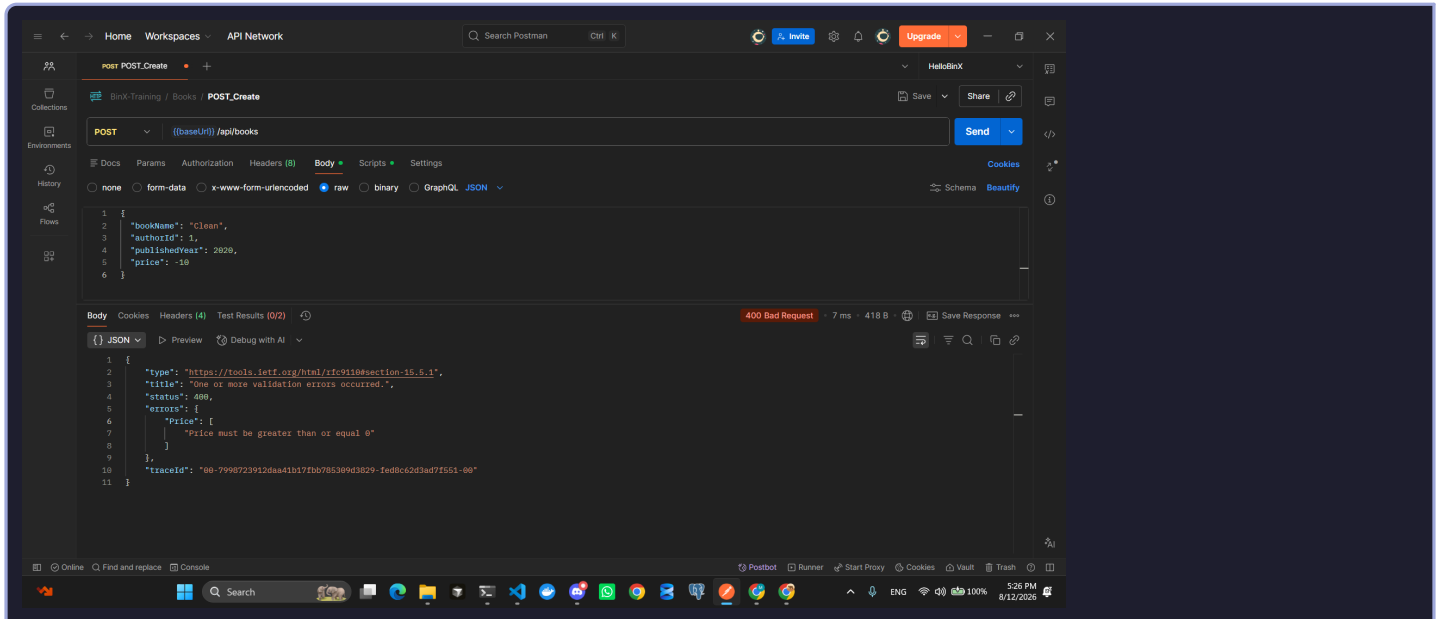
#### ■ Postman Screenshot



## Test Case 5 Invalid Price (negative value)

|                   |                                                                     |
|-------------------|---------------------------------------------------------------------|
| Description       | Sending a negative price to verify that price cannot be below zero. |
| Field Tested      | price                                                               |
| Input Value       | -10                                                                 |
| HTTP Status       | <b>400 Bad Request</b>                                              |
| Validation Result | Server returned: <b>"Price must be greater than or equal 0"</b> .   |

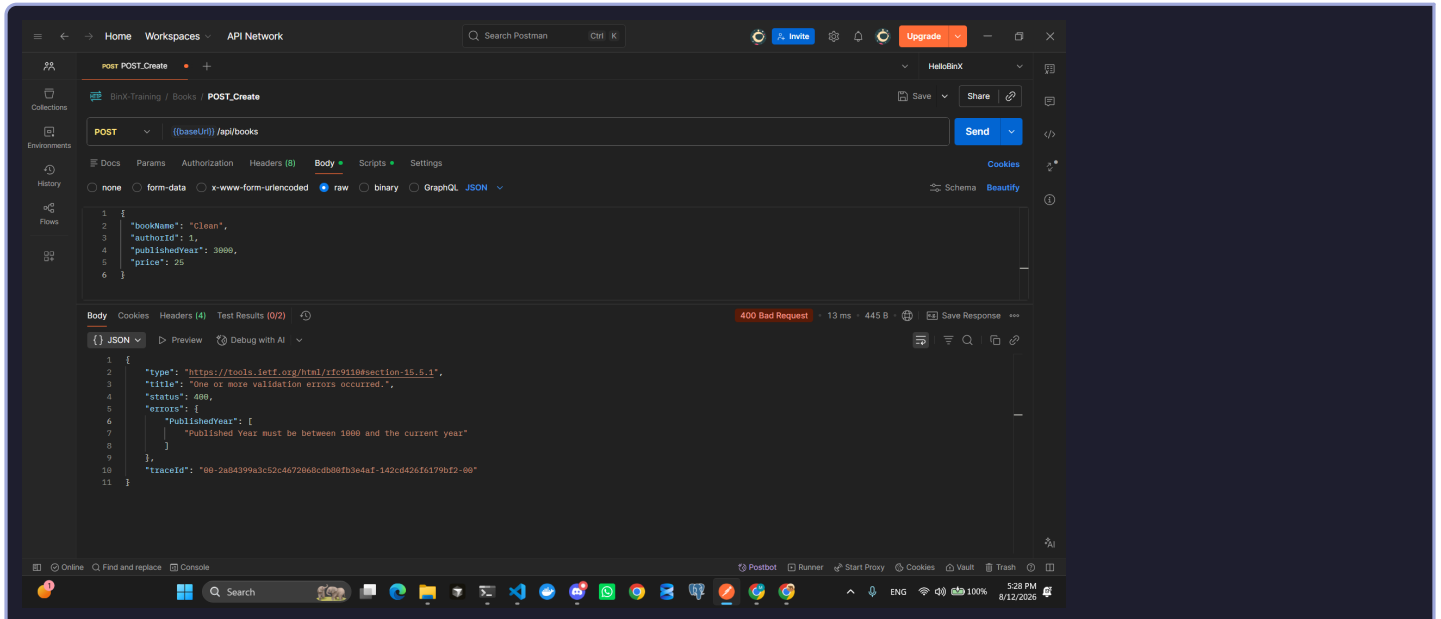
### ■ Postman Screenshot



## Test Case 6 Invalid Published Year (future year out of range)

|                   |                                                                                        |
|-------------------|----------------------------------------------------------------------------------------|
| Description       | Sending a year of 3000 to verify that published year must not exceed the current year. |
| Field Tested      | publishedYear                                                                          |
| Input Value       | 3000                                                                                   |
| HTTP Status       | <b>400 Bad Request</b>                                                                 |
| Validation Result | Server returned: <b>"Published Year must be between 1000 and the current year"</b> .   |

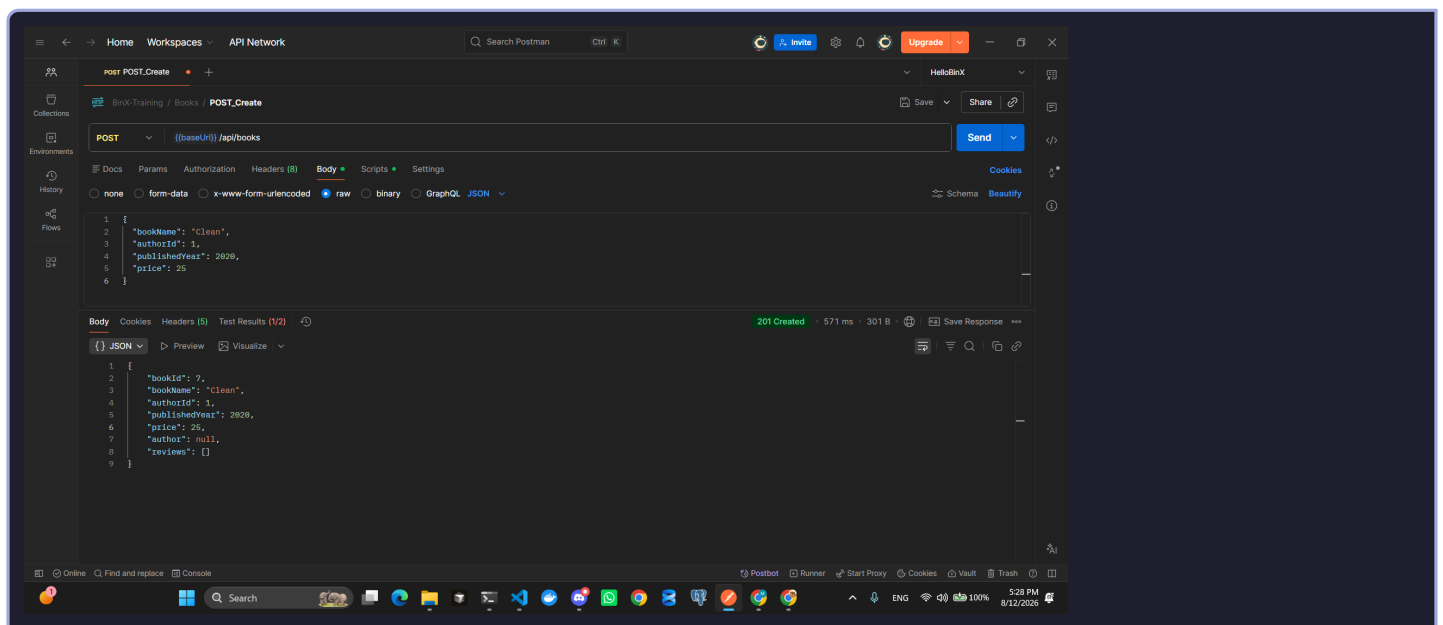
### ■ Postman Screenshot



## Test Case 7 Valid Create Request — Success

|                   |                                                                                                                                                                    |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description       | Sending a fully valid payload to confirm the book is created successfully.                                                                                         |
| Field Tested      | All fields                                                                                                                                                         |
| Input Value       | bookName: "Clean", authorId: 1, publishedYear: 2020, price: 25                                                                                                     |
| HTTP Status       | <b>201 Created</b>                                                                                                                                                 |
| Validation Result | Server returned <b>201 Created</b> with the new book object: bookId: 7, bookName: "Clean", authorId: 1, publishedYear: 2020, price: 25, author: null, reviews: []. |
| Note              | This is the happy-path case confirming all validations passed.                                                                                                     |

### ■ Postman Screenshot





# Section 2

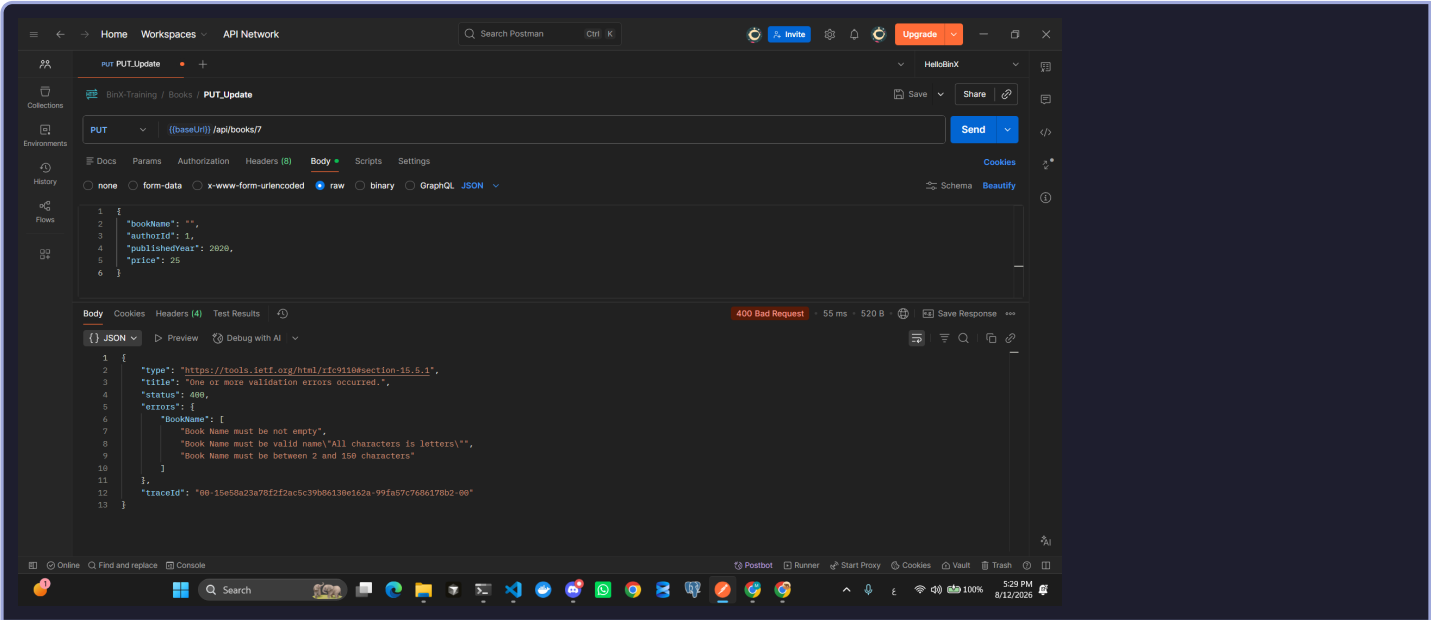
## Update Book Endpoint Validation

PUT /api/books/{id}

The following test cases validate the request body constraints enforced by the Update Book endpoint (PUT). The test target is book ID 7, created in Section 1.

| Test Case 1       | Empty Book Name                                                                                                                                                                            |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description       | Sending an empty string as book name in the PUT request to confirm the same required-field rules apply.                                                                                    |
| Field Tested      | bookName                                                                                                                                                                                   |
| Input Value       | "" (empty string)                                                                                                                                                                          |
| HTTP Status       | 400 Bad Request                                                                                                                                                                            |
| Validation Result | Server returned three validation errors: "Book Name must be not empty", "Book Name must be valid name\All characters is letters\\"", and "Book Name must be between 2 and 150 characters". |

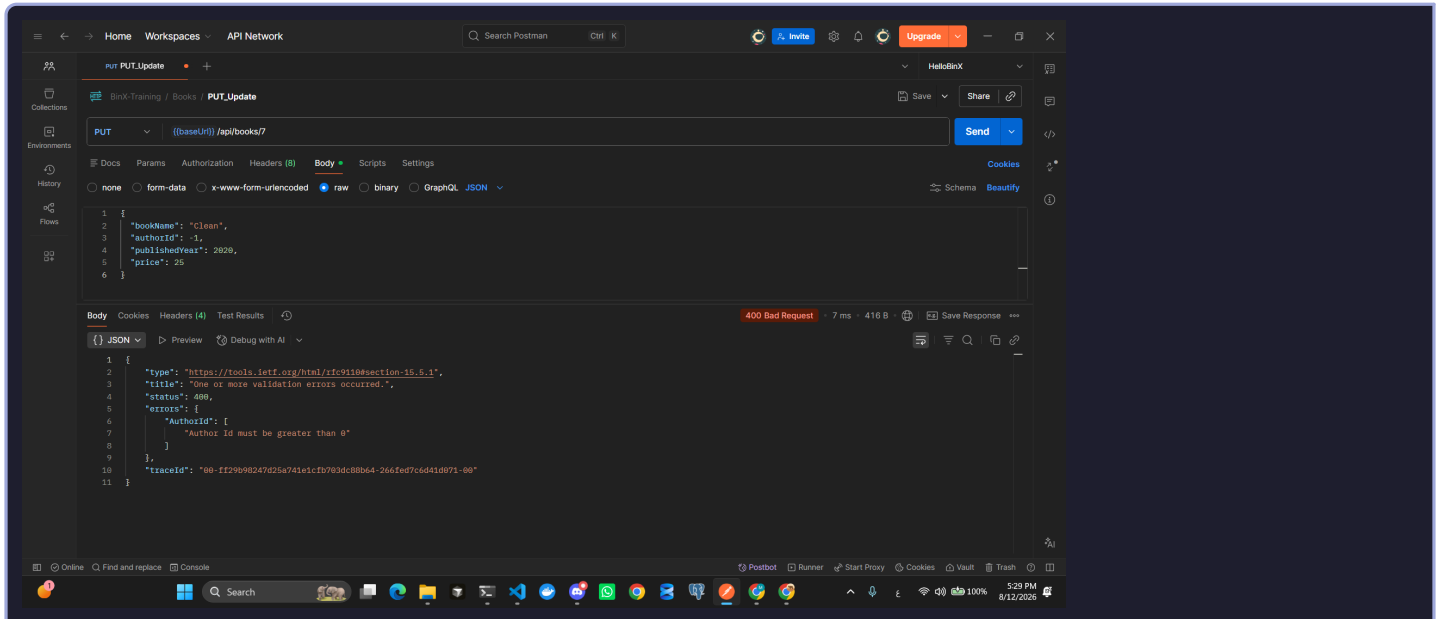
### Postman Screenshot



## Test Case 2 Invalid Author ID (negative value)

|                   |                                                                                      |
|-------------------|--------------------------------------------------------------------------------------|
| Description       | Sending a negative authorId in the PUT request to verify the positive-ID constraint. |
| Field Tested      | authorId                                                                             |
| Input Value       | -1                                                                                   |
| HTTP Status       | <b>400 Bad Request</b>                                                               |
| Validation Result | Server returned: "Author Id must be greater than 0".                                 |

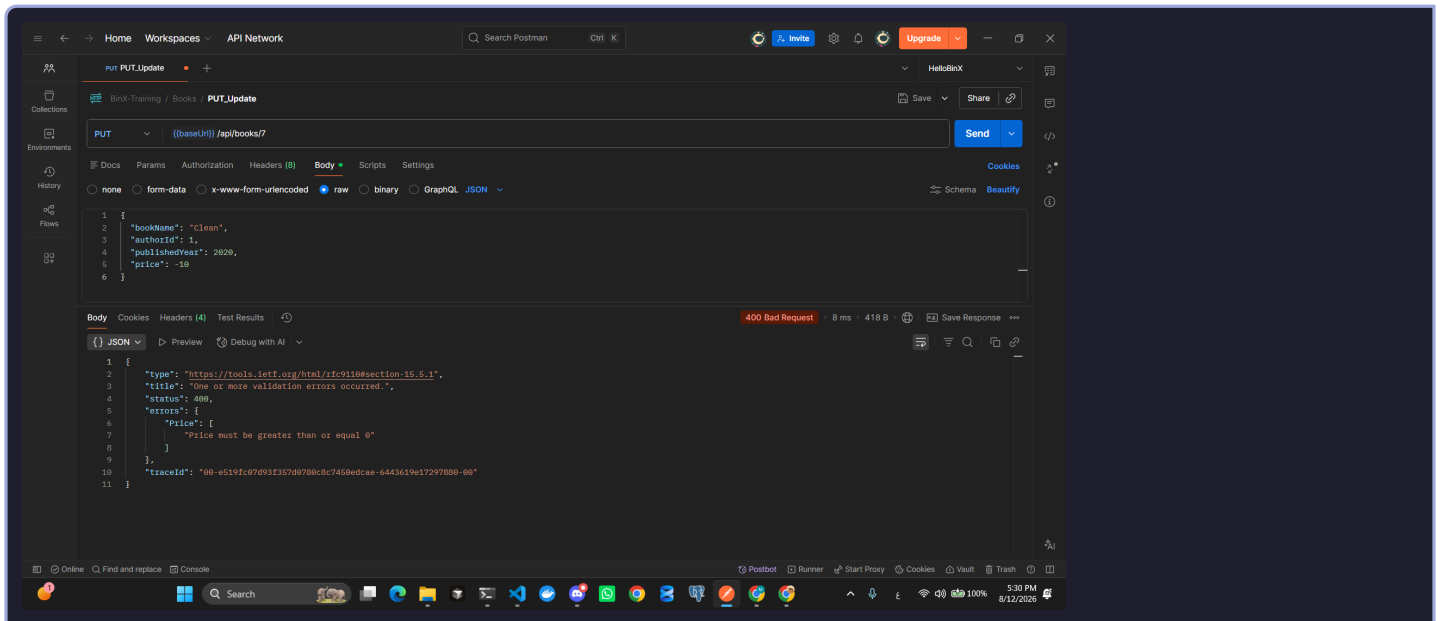
### ■ Postman Screenshot



### Test Case 3 Invalid Price (negative value)

|                   |                                                                                          |
|-------------------|------------------------------------------------------------------------------------------|
| Description       | Sending a negative price in the PUT request to verify the non-negative price constraint. |
| Field Tested      | price                                                                                    |
| Input Value       | -10                                                                                      |
| HTTP Status       | <b>400 Bad Request</b>                                                                   |
| Validation Result | Server returned: <b>"Price must be greater than or equal 0"</b> .                        |

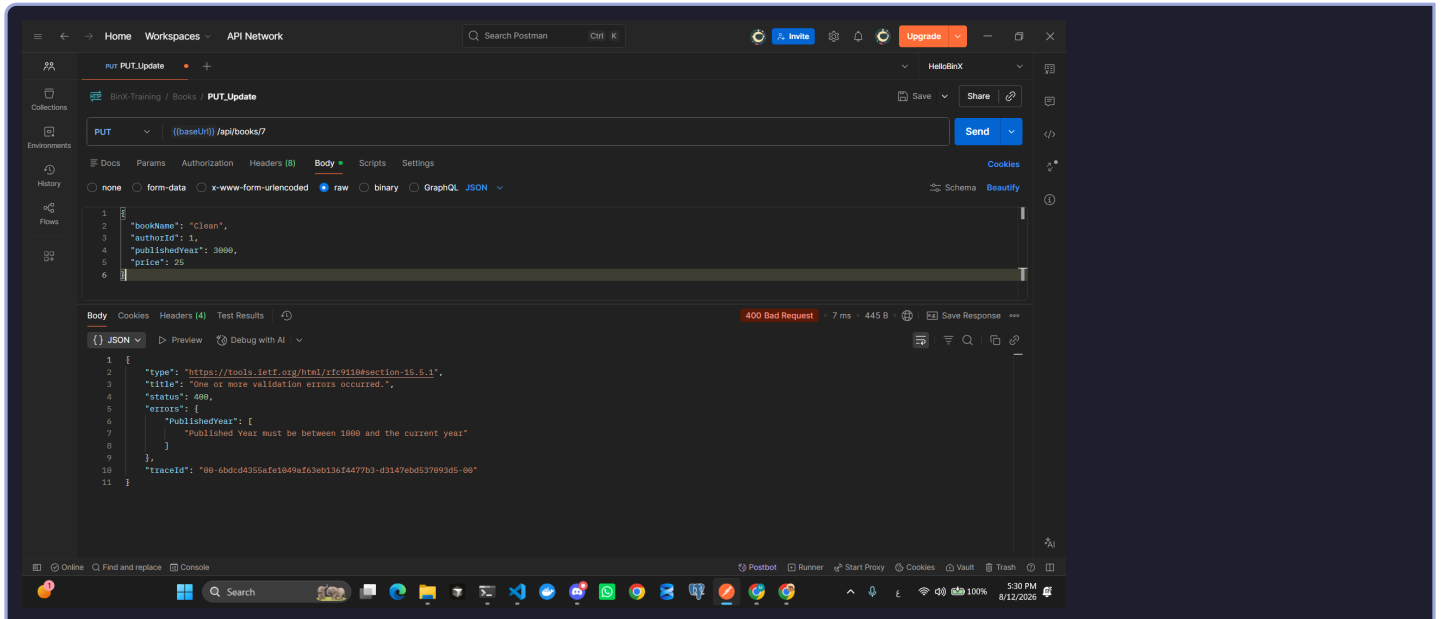
#### ■ Postman Screenshot



#### Test Case 4 Invalid Published Year (future year out of range)

|                   |                                                                                      |
|-------------------|--------------------------------------------------------------------------------------|
| Description       | Sending a year of 3000 in the PUT request to verify the year-range constraint.       |
| Field Tested      | publishedYear                                                                        |
| Input Value       | 3000                                                                                 |
| HTTP Status       | <b>400 Bad Request</b>                                                               |
| Validation Result | Server returned: <b>"Published Year must be between 1000 and the current year"</b> . |

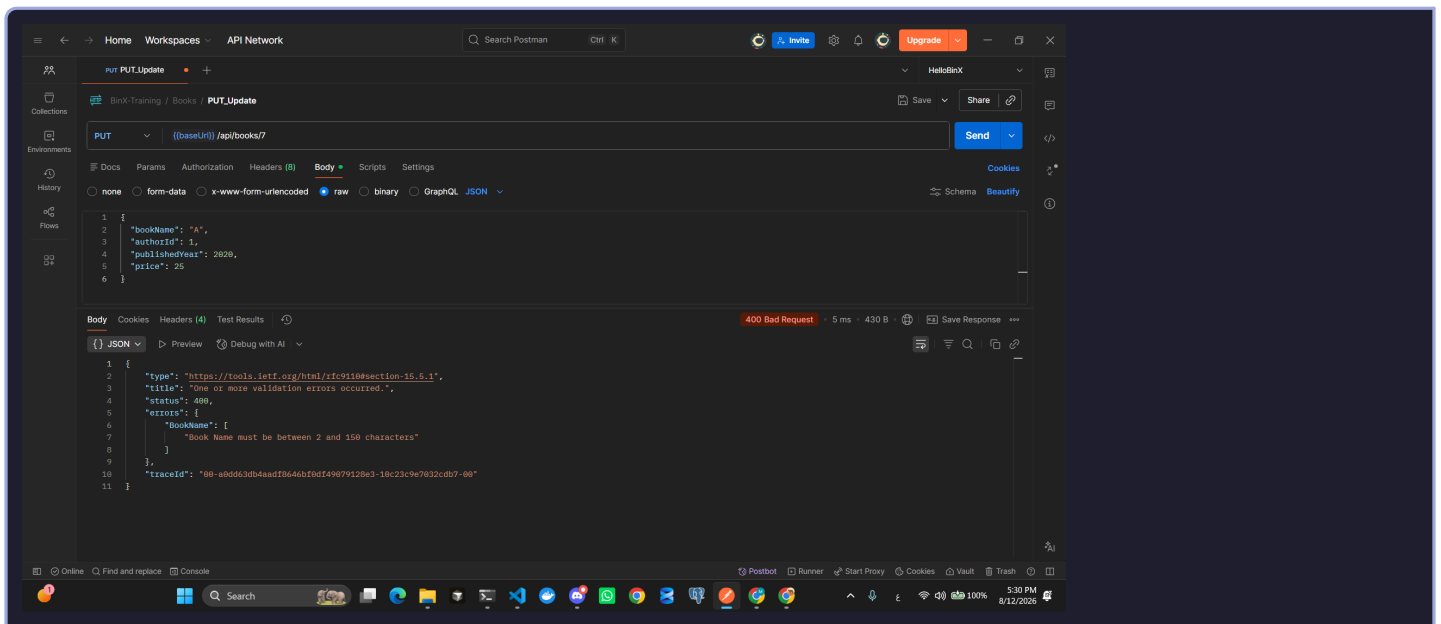
#### ■ Postman Screenshot



## Test Case 5 Book Name Too Short (1 character)

|                   |                                                                                            |
|-------------------|--------------------------------------------------------------------------------------------|
| Description       | Sending a single-character book name in the PUT request to verify the minimum-length rule. |
| Field Tested      | bookName                                                                                   |
| Input Value       | "A" (1 character)                                                                          |
| HTTP Status       | <b>400 Bad Request</b>                                                                     |
| Validation Result | Server returned: <b>"Book Name must be between 2 and 150 characters"</b> .                 |

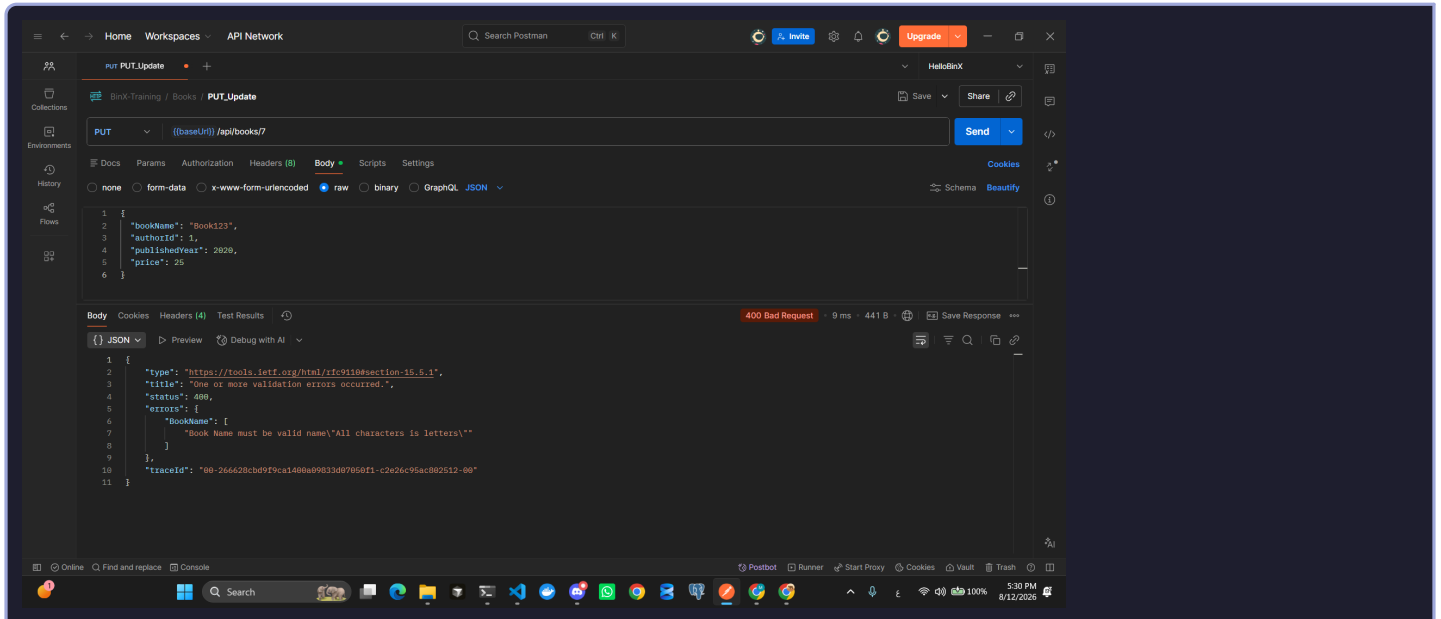
### ■ Postman Screenshot



## Test Case 6 Book Name Contains Invalid Characters

|                   |                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------|
| Description       | Sending a book name with digits in the PUT request to verify the letters-only constraint. |
| Field Tested      | bookName                                                                                  |
| Input Value       | "Book123" (contains digits)                                                               |
| HTTP Status       | <b>400 Bad Request</b>                                                                    |
| Validation Result | Server returned: <b>"Book Name must be valid name\All characters is letters\"</b> .       |

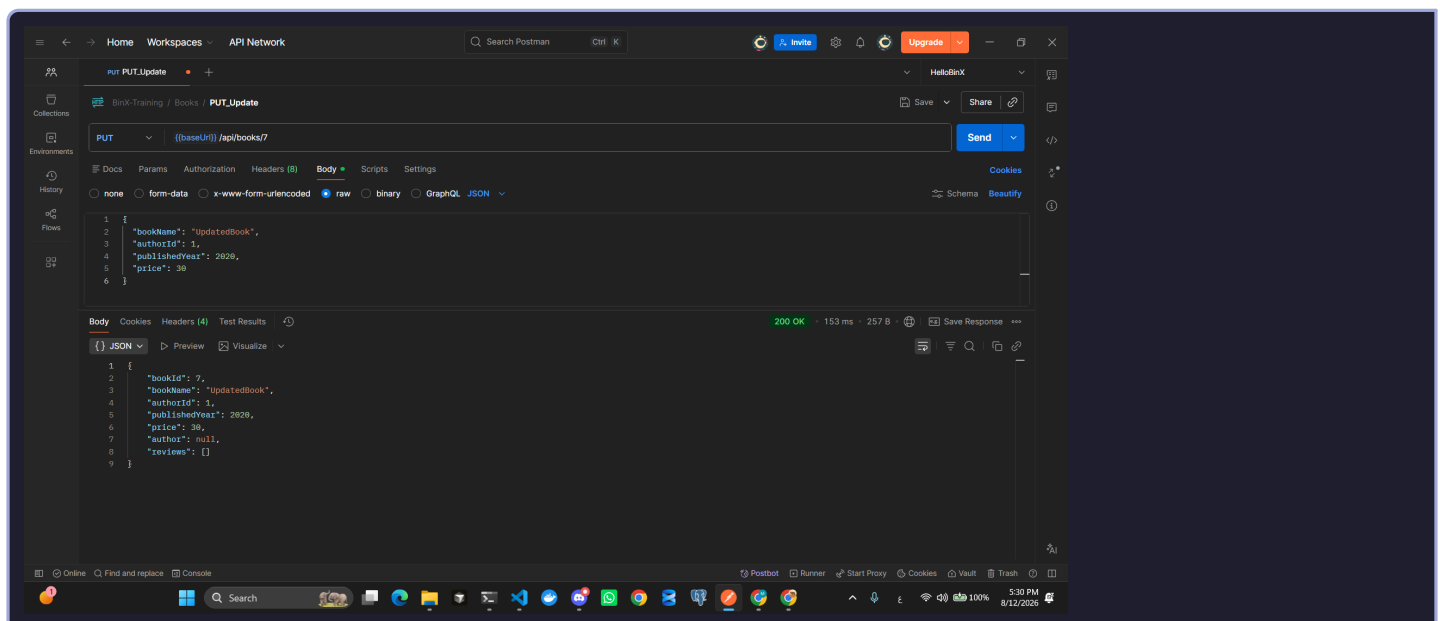
### ■ Postman Screenshot



## Test Case 7 Valid Update Request — Success

|                   |                                                                                                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description       | Sending a fully valid payload to confirm the book is updated successfully.                                                                                              |
| Field Tested      | All fields                                                                                                                                                              |
| Input Value       | bookName: "UpdatedBook", authorId: 1, publishedYear: 2020, price: 30                                                                                                    |
| HTTP Status       | 200 OK                                                                                                                                                                  |
| Validation Result | Server returned <b>200 OK</b> with the updated book object: bookId: 7, bookName: "UpdatedBook", authorId: 1, publishedYear: 2020, price: 30, author: null, reviews: []. |
| Note              | This is the happy-path case confirming the update operation succeeded.                                                                                                  |

### ■ Postman Screenshot



## Section 3 — Testing Summary

| Metric                    | Value | Details                                 |
|---------------------------|-------|-----------------------------------------|
| Create Book Tests         | 7     | 6 validation (400) + 1 success (201)    |
| Update Book Tests         | 7     | 6 validation (400) + 1 success (200)    |
| Total Test Cases          | 14    | 12 invalid + 2 valid                    |
| 400 Bad Request responses | 12    | All invalid requests correctly rejected |
| 201 Created response      | 1     | Valid POST /api/books succeeded         |
| 200 OK response           | 1     | Valid PUT /api/books/7 succeeded        |

### Validation Rules Confirmed

| Rule                  | Constraint                                          |
|-----------------------|-----------------------------------------------------|
| bookName — Required   | Must not be empty (400 if blank)                    |
| bookName — Length     | Must be between 2 and 150 characters                |
| bookName — Characters | All characters must be letters (no digits, symbols) |
| authorId — Range      | Must be greater than 0 (positive integer)           |
| price — Range         | Must be greater than or equal to 0                  |
| publishedYear — Range | Must be between 1000 and the current year           |
| Valid request — POST  | Returns 201 Created with full book object           |
| Valid request — PUT   | Returns 200 OK with updated book object             |

**Conclusion:** All 14 test cases executed successfully. The API correctly enforces validation on all constrained fields and returns descriptive 400 Bad Request responses for invalid inputs. Valid payloads produce the expected 201 Created (POST) and 200 OK (PUT) responses, confirming the endpoint implementation is working as intended.